

March 2026 Cold Box Data-taking status summary

Noise runs, 270 nm SPE-oriented scans, and 367 nm dynamic-range scans

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Requested scope vs. data categories taken

Requested in the March 6 plan

- ▶ Noise characterization with LED off: FFT, baseline STD, baseline offset.
- ▶ Low-light LED scans to set a working point around 1 average PE.
- ▶ VGAIN and bias scans to extract SPE response and SNR.
- ▶ High-light LED scans to probe dynamic range and saturation.

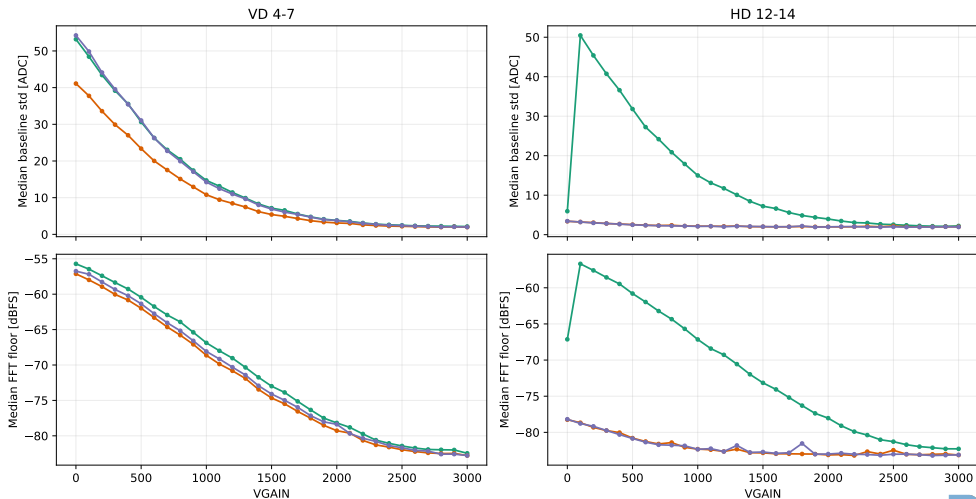
What we actually accumulated

- ▶ **Noise runs:** dark VGAIN scans plus oscilloscope captures under several electrical configurations.
- ▶ **270 nm scans:** low-light tuning scans and targeted VGAIN/bias matrices for SPE-oriented studies.
- ▶ **367 nm scans:** high-light scans used to push channels toward dynamic-range and saturation studies.
- ▶ **Coverage picture:** working assumption is that VD channels 0–7 and HD channels 8–15 are largely covered; the systematic campaign still missing is mainly SoF / AFE2, channels 16–23.

Nature of the noise runs

- ▶ These are **LED-off** runs meant to separate readout/electronics noise from LED and signal-shape effects.
- ▶ The main observables are **baseline STD** and **FFT floor / spectral structure** as a function of VGAIN.
- ▶ The run family includes standard dark scans, detector-connected variants, and cable/choke follow-ups.
- ▶ This is the dataset used to compare noise between measurements and to flag obvious outliers before interpreting SPE scans.

Mar 16 d13 dark Mar 19 d13 dark Mar 19 d13 no choke
Noise comparison across representative dark runs



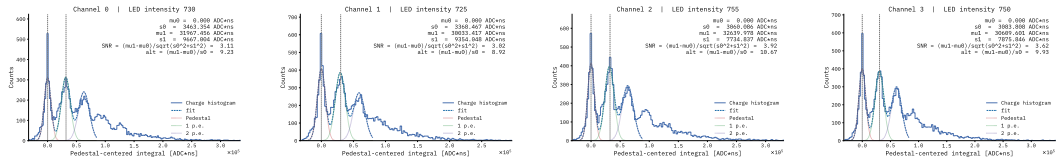
270 nm scans: purpose and workflow

- ▶ The 270 nm scans were used as the **SPE-oriented working dataset**.
- ▶ First step: scan LED intensity channel-by-channel to find a usable low-light operating point with visible pedestal / first-PE structure.
- ▶ Second step: with those fixed intensities, run **VGAIN scans** and **VGAIN-bias matrices** to map SPE response, SNR trends, and the dependence on front-end settings.
- ▶ In practice, this produced:
 - ▶ VD tuning scans for channels 0–7.
 - ▶ HD tuning scans and dedicated matrices for channels 12–15.
 - ▶ Partial / resumed 2D matrices for VD follow-up studies.

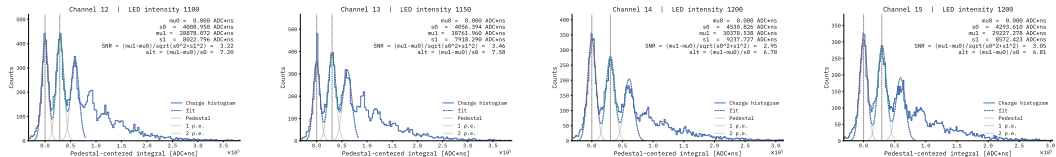
Representative data	Purpose
20260317_daphne-13_led_vgain_bias_HDscan	HD matrix, 270 nm, fixed LED per channel group, VGAIN and AFE1 bias scan for SPE-oriented evaluation
20260319_vd_ch0to3_led270_*	VD channel tuning and retakes to narrow the SPE working point
20260320_vd_ch0to3_led270_0700_070000	High-statistics retake around the selected VD working region

Fingerplot examples (270 nm, low light)

VD examples from the most recent low-light analysis, shown with slightly higher LED settings (channels 0–3).



HD examples kept from the current low-light runs (channels 12–15).



367 nm scans: dynamic range and saturation

- ▶ The 367 nm scans were **not** used to set the SPE working point.
- ▶ They were used as the **high-light dataset** to drive the system toward dynamic-range and saturation studies.
- ▶ In practice, the 367 nm scans explored wide LED settings at fixed FE configurations, mainly to test when the response leaves the linear / low-occupancy regime.
- ▶ This complements the 270 nm low-light scans: 270 nm for SPE-oriented operation, 367 nm for high-occupancy / saturation behavior.

Representative HD examples from the March 18 remote 367 nm scan.

Current dataset status

- ▶ **Noise / no LED:** the dataset is broad and includes multiple electrical configurations, dark VGAIN scans, and oscilloscope captures.
- ▶ **VD low-light scans:** channels 0–3 and 4–7 have dedicated 270 nm working-point scans and retakes.
- ▶ **HD low-light scans:** channels 12–15 have explicit fingerplot examples and a dedicated 270 nm VGAIN-bias matrix.
- ▶ **High-light scans:** 367 nm scans exist and were used for dynamic-range / saturation studies.
- ▶ **Remaining systematic gap:** the main channel family still awaiting a comparable dedicated campaign is SoF / AFE2, channels 16–23.

Practical interpretation

- ▶ We already have a useful dataset for a first CE/WE performance discussion.
- ▶ The next priority is not more depth on already-covered VD/HD examples, but systematic completion of the SoF channel family.